

Institute for Energy Technology (IFE)
Instituttveien 18, 2007 Kjeller, Norway

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Editorial Office
Computer Methods in Applied Mechanics and Engineering

Dear Editor,

We submit **“PDE-Agents: An LLM-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning”** for consideration as an original research article in CMAME.

The manuscript introduces a computational framework in which three specialist large language model agents autonomously set up, run, analyse, and iteratively improve finite element simulations from natural-language descriptions. A GraphRAG knowledge base (Neo4j, 768-dim vector embeddings) grounds agent reasoning in curated material properties, failure patterns, and prior run lineage. We propose a novel *KG Smart* integration strategy and show that *integration pattern*, rather than knowledge content alone, determines whether retrieval augmentation improves agent performance.

The contribution is methodological. We present and evaluate a new computational framework rather than apply an existing solver to a new problem. The evaluation includes verification and validation confirming $\mathcal{O}(h^2)$ convergence, a three-way ablation across 50 tasks, a novel-material experiment (MPF = 1.00 vs. 0.34 for the baseline), and production metrics from 1,369 runs (97.8% success). All code and data are available at <https://github.com/MatPro-IFE/pde-agents>.

This work falls within CMAME’s scope of physically based machine-learning methods and computational automation. To our knowledge, no prior work combines multi-agent LLM orchestration with production FEM solvers and graph-based retrieval for computational mechanics. The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission.

Sincerely,

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